

Trade Liberalization and Local Labor Market Adjustment: New Evidence from Korea

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Abstract

Utilizing the launch of the China–Korea free trade agreement (FTA) as an exogenous event, this study analyzes the impact of increased trade with China on local labor markets in Korea. In particular, we employ the method developed by Pierce and Schott (2016, 2020), who calculate changes in tariff rates for each region to analyze the effects of trade flows on local labor markets. While the FTA does not significantly affect overall employment, we find compositional shifts: employment among male, routine, and less-educated workers significantly increases in regions more exposed to import tariff reductions. However, we do not find statistically significant changes in employment by job type.

Keywords: Trade liberalization, local labor markets, Korea

JEL Classification: J20, J23, R23

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Abbreviations: FTA, free trade agreement; LALFS, Local Area Labor Force Survey; LZ, living zone

1 Introduction

Since China’s accession to the World Trade Organization in 2001, global trade has increased substantially. As a result, numerous studies have investigated the effects of this change on the economy as a whole, particularly the labor market. However, no clear consensus exists on how trade affects the labor market. For instance, Autor et al. (2013), Acemoglu et al. (2016), and Pierce and Schott (2016) found that trade competition with China had a significant impact on the labor market, resulting in a decline in the number of manufacturing workers in the United States. Meanwhile, Feenstra et al. (2019) and Feenstra and Sasahara (2018) reported conflicting results, suggesting that trade with China positively impacted both the United States and East Asian economies. Furthermore, Aghion et al. (2024) showed that exposure to output trade competition had a negative impact on the employment of French firms, particularly for low-productivity firms. In contrast, the input component of trade may have had a positive effect, particularly for firms that import a significant portion of their inputs from China and benefit from cheaper input goods.

This paper aims to contribute to the literature by providing empirical evidence: We examine the extent to which trade liberalization with China affected the Korean labor market. In particular, we use the China–Korea free trade agreement (henceforth FTA), which entered into force in December 2015, as an exogenous event to analyze the effect of increased trade with China on the Korean labor market. Since China opened its markets in the early 2000s, trade with China has steadily increased, and Korea has become an important trading partner, ranking second among China’s trading partners by 2022 (Korea International Trade Association 2023).

The fact that many trade-related factors can influence employment fluctuations makes the empirical analysis difficult. For example, Korea implemented FTAs with Chile in 2004, Singapore in 2006, the Association of Southeast Asian Nations in 2007, India in 2010, and the United States in 2012 (with revised negotiations in 2019) during the same broad period in which trade with China increased. Additionally, other changes in the labor market during the sample period could have affected the outcome variables of interest.⁵

Therefore, to isolate the effect of China’s trade liberalization on the Korean labor market, we treat the China–Korea FTA as an exogenous event and adopt the approach of Pierce and Schott (2016) and Pierce and Schott (2020) by calculating overall changes in tariff rates for each region. Using the Local Area Labor Force Survey (LALFS) data provided by Statistics

⁵For example, the mandatory retirement age in Korea increased from 55 to 60 years in 2016 (for firms employing more than 300 workers) and 2017 (for firms employing fewer than 300 workers), which could have an impact on employment and wages.

Korea, we then apply the difference-in-differences method to exploit regional variation in trade exposure and estimate the causal effect of trade liberalization with China on local labor markets in Korea.

The summary of the results is as follows. First, we do not find a statistically significant effect of the China–Korea FTA on overall employment in regions more exposed to import tariff reductions. Second, male employment increases throughout the post-FTA period, whereas the estimates for female employment are negative but imprecise. Third, routine employment increases, particularly in agricultural, craft, and equipment-related occupations, while manual employment decreases. Fourth, the estimates do not show statistically significant changes in overall regular or temporary employment. By education, less-educated employment increases while highly-educated employment decreases, although some subgroup estimates require caution because of pre-FTA differences.

This is not the first study to evaluate the effect of trade with China on the Korean labor market. Kim (2006) and Kim and Choi (2015) found that trade expansion with China had a positive impact on employment in Korea. Additionally, Kim and Shikher (2015) used a computable general equilibrium model to predict that the effect of the FTA on employment would vary by industry, with an increase in employment expected in the manufacturing sector but a decrease in the food sector. In contrast, Koo and Whang (2018) focused only on the manufacturing sector and demonstrated that job losses resulting from the China shock were offset by job creation due to the same shock. To address potential confounding factors between trade and employment, we introduce an exogenous event, the FTA, and leverage regional variation, which distinguishes our study from previous research.

The remaining sections are organized as follows. Section 2 introduces the data and variables constructed for the analysis and presents the empirical strategy. Section 3 presents the results. Finally, Section 4 concludes.

2 Data and empirical analysis

2.1 Data and variables

For the empirical analysis, we use LALFS data from 2013 to 2019 provided by Statistics Korea. The LALFS, which began in 2008, stands as the sole domestic dataset offering employment data on industries and occupations across 162 cities and counties in Korea. Consequently, it serves as an appropriate resource for our research, which leverages regional variation to identify the coefficient of interest. The LALFS is a nationally representative, biannual survey of household members aged 15 years and older. We use the survey’s city-

and county-level weights to construct regional employment totals and employment shares and report employment outcomes in thousands of workers. We use data from 2013 (rather than the initial period of 2008) in the analysis because of a major change in sample design between 2012 and 2013, resulting in a sharp increase in the number of workers in 2013.⁶

As the China–Korea FTA entered into force in December 2015, the years from 2013 to 2015 and from 2016 to 2019 are considered the pre-FTA and post-FTA periods, respectively. In the benchmark analysis, we use 162 cities and counties as regional units, include living-zone-by-year fixed effects, and cluster standard errors at the living-zone level (56 units), as defined by the Statistics Research Institute of Statistics Korea.⁷

Following Pierce and Schott (2016) and Pierce and Schott (2020), we then construct variables that quantify the extent to which the FTA disproportionately affects regions. First, the difference between the most-favored-nation tariff rate (henceforth MFN rate) and the post-FTA tariff rate (henceforth FTA rate) for each industry is calculated and used as a proxy for the FTA’s impact. The following formula is used to construct the variable:

$$FTAGAP_i = MFNrate_i - FTArate_i, \quad (1)$$

where i represents each industry. MFN and FTA rates are obtained from Korea’s tariff schedule, and $FTAGAP_i$ is the average product-level difference within each two-digit industry in 2016 (see Table A.1). The tariff reduction was zero for some industries and reached approximately 4.8 percentage points in the most exposed industry. Although the absolute size of the change is not large, $FTAGAP_i$ captures variation among industries.

Next, using the difference between MFN and FTA rates, we construct $FTAGAP_r$ as an employment-share-weighted average across industries for each region:

$$FTAGAP_r = \sum_i \frac{L_{ir,2015}}{L_{r,2015}} FTAGAP_i, \quad (2)$$

where r denotes a region, defined either at the city/county level (162 units) or as a living zone (56 units), and L denotes employment estimated using the LALFS city- and county-level weights. Because the China–Korea FTA entered into force in December 2015, 2015 is the

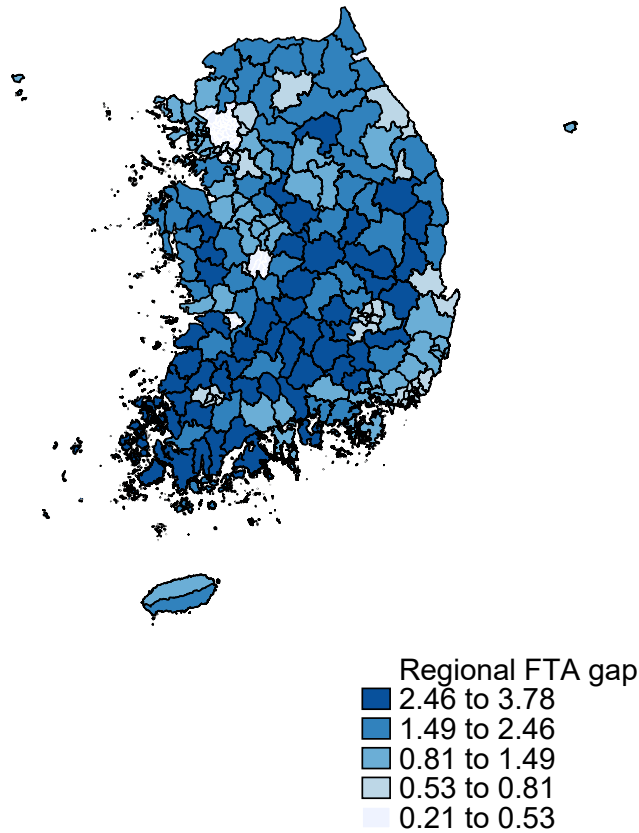
⁶Prior to 2012, the survey was conducted four times a year (i.e., quarterly), but since 2013, it has been conducted twice a year (April and October). Additionally, the survey sample increased from 174,000 households in the first quarter of 2012 to 199,000 households after 2013. As of 2020, approximately 234,000 households are included in the survey (Statistics Korea). To eliminate seasonal fluctuations, we use only the October sample from each year’s survey in this analysis.

⁷Living zones (LZs) are comparable to commuting zones in the United States. They differ from regional administrative units in that they capture residents’ actual living areas. The LZ classification consists of 56 zones that encompass all local governments and regions in Korea: central-city living zones (22 units), urban–rural living zones (13 units), and rural living zones (21 units).

baseline year. $FTAGAP_r$ then measures each region's exposure to tariff reductions under the China–Korea FTA.

Figure 1 maps each region's $FTAGAP_r$ value, calculated using Equation (2). Darker-colored regions have greater exposure to the China–Korea FTA. For example, Jeollanam-do (southwest) and Gyeongsangbuk-do (central east) are relatively exposed because the tariff reduction on agricultural products, which are produced extensively in these two regions, was approximately 3.0 percentage points.

Figure 1. Regional exposure to import tariff reductions under the China–Korea FTA



Notes: The map reports $FTAGAP_r$, the average of 2016 industry-level import tariff reductions using 2015 survey-weighted employment shares. Darker shades indicate greater exposure.

2.2 Empirical strategy

We employ a difference-in-differences method to estimate the effect of the China–Korea FTA on local labor markets. For the main analysis, we use the following specification:

$$Y_{rt} = \alpha_r + \alpha_t + \alpha_{lt} + \sum_t \beta_t \mathbf{1}\{year = t\} \times FTAGAP_r + X'_{rt} \gamma + \varepsilon_{rt}, \quad (3)$$

where α_r is a region fixed effect that captures unique regional characteristics, and α_t is a year fixed effect that captures time-specific shocks affecting all regions (e.g., economic fluctuations). α_{lt} denotes living-zone-by-year fixed effects, and $\mathbf{1}\{year = t\}$ is an indicator equal to 1 in year t . The dependent variables are survey-weighted employment measures, expressed in thousands of workers and disaggregated by gender, occupation, job type (i.e., regular and temporary), and education level to investigate the heterogeneous effects of the China–Korea FTA.⁸ The coefficient of interest is β_t : if β_t is less (resp. greater) than zero, the effect of the China–Korea FTA on employment is negative (resp. positive). Statistical significance is assessed using the clustered standard errors. Following Pierce and Schott (2020), X_{rt} includes the contemporaneous FTA tariff level, the export-side FTA gap, and year interactions with the region’s 2013 routine- and less-educated-employment shares. We cluster the error term at the living-zone level.

Before presenting the regression results, we compute average regional employment levels for each group before and after the implementation of the FTA (columns 1 and 2 of Table 1). Column 3 reports the full-period average, while the last column reports the before-minus-after difference and its statistical significance based on a t -test.

⁸Observations with missing education are excluded only from the education-specific outcomes, so the two education groups do not sum exactly to total employment.

Table 1. Average employment levels before and after the China–Korea FTA

Variable	Before FTA		After FTA		Overall		t-test
	N	Mean	N	Mean	N	Mean	Difference
Employment	486	153.551 [21.072]	648	160.174 [18.665]	1134	157.336 [13.969]	−6.623
Male	486	90.942 [12.270]	648	93.479 [10.594]	1134	92.392 [8.015]	−2.537
Female	486	62.609 [8.813]	648	66.695 [8.083]	1134	64.944 [5.964]	−4.086
Managerial occ.	486	1.761 [0.350]	648	1.444 [0.215]	1134	1.580 [0.194]	0.317
Professional occ.	486	30.758 [5.755]	648	33.658 [5.439]	1134	32.415 [3.966]	−2.899
Clerical occ.	486	26.851 [4.827]	648	28.933 [4.344]	1134	28.041 [3.230]	−2.082
Sales occ.	486	17.548 [2.909]	648	17.051 [2.335]	1134	17.264 [1.825]	0.497
Agricultural occ.	486	11.151 [0.306]	648	10.212 [0.248]	1134	10.614 [0.194]	0.939**
Craft occ.	486	14.095 [1.786]	648	14.418 [1.486]	1134	14.280 [1.143]	−0.322
Equipment occ.	486	18.711 [1.704]	648	18.867 [1.449]	1134	18.800 [1.104]	−0.156
Service occ.	486	15.258 [2.082]	648	17.061 [1.916]	1134	16.288 [1.412]	−1.803
Elementary occ.	486	17.417 [2.082]	648	18.530 [1.885]	1134	18.053 [1.398]	−1.114
Regular worker	486	79.278 [12.317]	648	87.974 [11.545]	1134	84.247 [8.447]	−8.696
Temporary worker	486	22.853 [3.265]	648	22.509 [2.768]	1134	22.657 [2.111]	0.344
Highly educated	486	53.245 [10.106]	648	60.003 [9.490]	1134	57.107 [6.938]	−6.758
Less educated	486	98.053 [11.068]	648	98.356 [9.300]	1134	98.226 [7.120]	−0.303

Notes: Employment measures are population estimates, in thousands of workers, constructed using the LALFS city- and county-level weights. Brackets report standard errors of the mean. The last column reports the before-minus-after difference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 1 shows similar average employment levels before and after the FTA for most groups; the before–after difference is statistically significant only for agricultural workers. The overall mean in column 3 is 157.336 thousand workers, indicating that a city or county

had approximately 157,336 employed workers per year on average. The descriptive differences in this table should not be interpreted as causal effects of the FTA.

3 Empirical findings

3.1 Effect on the labor market as a whole and by gender

Table 2 presents the primary result of our study, emphasizing the impact of the China–Korea FTA on the local labor market as a whole and by gender. Model 1 (columns 1–3) employs region and year fixed effects, whereas Model 2 (columns 4–6) also includes living-zone-by-year fixed effects and serves as our baseline analysis. Both models control for the contemporaneous FTA tariff level, the export-side FTA gap, and year interactions with 2013 routine- and less-educated-employment shares. Because the difference-in-differences approach requires a parallel-trends assumption, we examine whether employment gaps between regions were stable before the FTA entered into force. Accordingly, Table 2 reports the estimated coefficients on the interactions between the year indicators and $FTAGAP_r$ in Equation (3). The estimate for 2015 is omitted because it is the base year.

Table 2. Employment effect of the China–Korea FTA

	Model 1			Model 2		
	(1) Overall	(2) Male	(3) Female	(4) Overall	(5) Male	(6) Female
2013	4.003*** (1.272)	1.905** (0.786)	2.098*** (0.686)	1.831 (1.870)	0.760 (1.344)	1.071 (0.796)
2014	1.696** (0.741)	0.820 (0.629)	0.876** (0.435)	1.871 (1.244)	0.772 (0.783)	1.099 (0.700)
2016	24.060*** (6.004)	18.803*** (6.617)	5.257 (5.307)	4.721 (4.806)	17.026* (8.511)	−12.306 (9.269)
2017	25.520*** (7.193)	20.376*** (6.915)	5.143 (5.837)	3.450 (6.358)	18.216** (9.038)	−14.767 (10.312)
2018	24.931*** (8.236)	22.071*** (7.556)	2.860 (6.860)	3.283 (7.356)	20.623** (10.176)	−17.339 (12.300)
2019	24.416*** (8.750)	22.843*** (7.722)	1.574 (7.656)	−0.653 (8.406)	19.866* (11.335)	−20.519 (13.982)
Constant	128.848*** (14.300)	63.111*** (16.696)	65.738*** (7.045)	138.331*** (14.725)	57.130*** (20.390)	81.201*** (8.411)
N	1134	1134	1134	1134	1134	1134
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
LZ $\times$ year FE	No	No	No	Yes	Yes	Yes
R-squared	1.000	1.000	0.999	1.000	1.000	0.999

Notes: Employment outcomes are population estimates, in thousands of workers, constructed using the LALFS city- and county-level weights. Standard errors clustered at the living-zone level are in parentheses. All specifications control for the contemporaneous FTA tariff level, the export-side FTA gap, and year interactions with 2013 routine- and less-educated-employment shares. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In Table 2, the post-FTA estimates for overall employment are small relative to their standard errors and are not statistically significant. We can therefore find no evidence that supports the usual notion that greater exposure to import tariff reductions changed total regional employment.

Because female and male workers may respond differently to trade liberalization (Besedeš et al., 2021), we separate the overall employment variable into male and female workers (columns 2, 3, 5, and 6 of Table 2). Male employment increases in every post-FTA year and is statistically significant at conventional levels. Female employment has negative point estimates throughout the post-FTA period, although none is statistically significant. These offsetting estimates help explain the absence of a detectable overall employment effect.

Figure 2 illustrates these changes in employment by plotting the coefficients of the interaction terms between year dummies and $FTAGAP_t$ with 90% confidence intervals. The

post-FTA estimates are positive for male employment and negative but less precise for female employment, while the overall estimates remain close to zero.

Figure 2. Effect of the China–Korea FTA on overall employment and by gender



Notes: Employment outcomes are population estimates, in thousands of workers, constructed using LALFS city- and county-level weights. Points are coefficient estimates from the baseline specification; vertical bars show 90% confidence intervals. The omitted year is 2015.

3.2 By occupation

In this subsection, we divide workers into several subgroups to study whether the effects of the FTA differed by occupation (see Autor et al. (2015) for further discussion of the heterogeneous effects of trade across occupations). We first classify workers into nine occupational groups following Acemoglu and Autor (2011) and Shim and Yang (2018) and then analyze the differential effects, if any, of the China–Korea FTA. Table 3 reports the results for all nine occupational groups, while Figure 3 plots the coefficients of interest after aggregating the nine groups into three broad categories: cognitive, routine, and manual workers.⁹

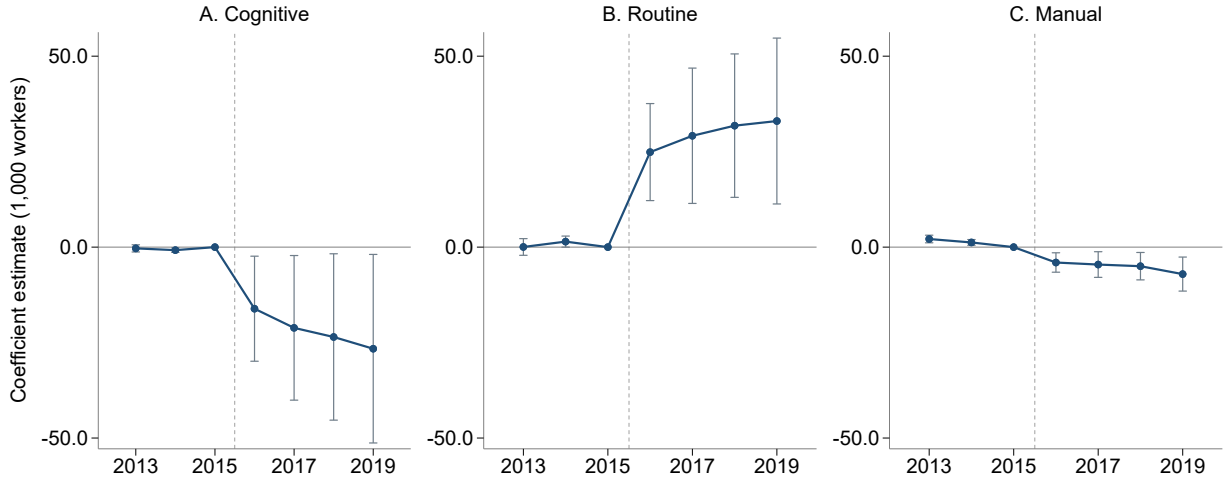
⁹Following Park et al. (2023), we classify managers, professionals, and related workers as cognitive workers; clerical and sales workers, skilled agricultural, forestry, and fishery workers, craft and related trades workers, and equipment, machine-operating, and assembling workers as routine workers; and service and elementary workers as manual workers.

Table 3. Effect of the China–Korea FTA on employment by occupation

	Cognitive			Routine			Manual		
	(1) Managerial	(2) Professional	(3) Clerical	(4) Sales	(5) Agricultural	(6) Craft	(7) Equipment	(8) Service	(9) Elementary
2013	0.027 (0.209)	−0.359 (0.460)	−0.140 (0.953)	0.284 (0.279)	−0.919*** (0.343)	0.287 (0.440)	0.527 (0.392)	1.409*** (0.313)	0.715* (0.395)
2014	−0.084 (0.095)	−0.710** (0.332)	0.765 (0.496)	0.100 (0.255)	−0.590* (0.317)	1.051** (0.478)	0.115 (0.403)	0.441 (0.273)	0.783* (0.409)
2016	4.106 (2.525)	−20.239* (10.768)	−6.483* (3.500)	8.686 (5.318)	8.529*** (2.490)	7.653* (4.066)	6.498** (2.911)	−3.992*** (1.288)	−0.038 (1.049)
2017	4.472 (2.763)	−25.619* (14.159)	−7.324** (3.338)	10.604 (6.944)	9.544*** (2.867)	9.542* (5.170)	6.792** (3.200)	−5.681** (2.242)	1.120 (1.698)
2018	4.807 (2.915)	−28.341* (16.064)	−7.990* (4.117)	10.977 (7.556)	10.064*** (3.143)	10.848* (5.821)	7.912** (3.653)	−5.780*** (2.097)	0.786 (1.850)
2019	5.057 (3.085)	−31.665* (17.986)	−9.611* (4.975)	12.434 (8.664)	10.385*** (3.400)	11.550* (6.568)	8.260* (4.497)	−6.523** (2.517)	−0.540 (1.756)
Constant	−4.182 (2.716)	61.487*** (12.293)	43.771*** (4.722)	1.385 (8.048)	−2.090 (3.423)	2.707 (5.535)	6.569 (6.509)	22.114*** (2.612)	6.570* (3.782)
N	1134	1134	1134	1134	1134	1134	1134	1134	1134
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
LZ × year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.960	0.998	0.999	0.998	0.965	0.998	0.998	0.999	0.998

Notes: Employment outcomes are population estimates, in thousands of workers, constructed using the LALFS city- and county-level weights. Standard errors clustered at the living-zone level are in parentheses. All specifications control for the contemporaneous FTA tariff level, the export-side FTA gap, and year interactions with 2013 routine- and less-educated-employment shares. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 3. Effect of the China–Korea FTA on employment by occupation



Notes: Employment outcomes are population estimates, in thousands of workers, constructed using LALFS city- and county-level weights. Points are coefficient estimates from the baseline specification; vertical bars show 90% confidence intervals. The omitted year is 2015.

The key findings can be summarized as follows. Routine employment increases significantly in every post-FTA year, while manual employment decreases significantly. Cognitive employment also has negative post-FTA estimates that are marginally significant. Table 3 shows the estimates by occupational group. Although some subgroup estimates require caution because of pre-FTA trends, agricultural and craft employment increase after the FTA, while employment in clerical and service occupations decreases. Estimates for professional employment are negative and marginally significant.

3.3 By job type

In this subsection, we analyze whether the China–Korea FTA had different effects on employment based on the degree of job security, given Korea’s pronounced dual labor market structure. Specifically, regular workers typically receive higher wages and enjoy greater job stability than temporary workers, making regular positions more desirable. Consequently, we examine whether the effects differed between regular and temporary workers. Table 4 presents the results, and Figure 4 shows the corresponding coefficients.

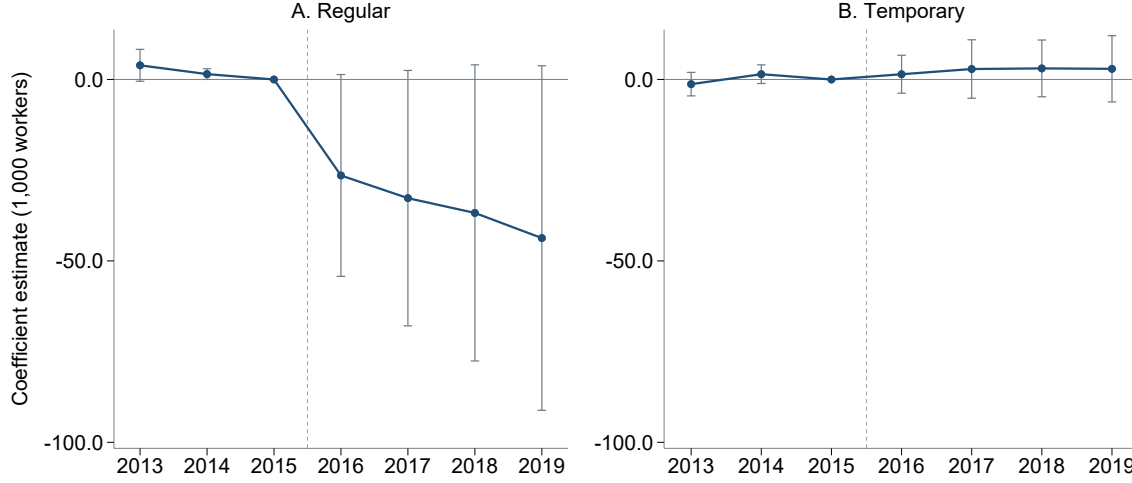
Table 4 does not show statistically significant post-FTA changes in overall regular or temporary employment. However, the estimates for female regular employment are negative and marginally significant. Overall, the results do not provide robust evidence that the China–Korea FTA changed employment differently by job type.

Table 4. Effect of the China–Korea FTA on employment by job type

	Regular			Temporary		
	(1) Overall	(2) Male	(3) Female	(4) Overall	(5) Male	(6) Female
2013	3.902 (2.671)	2.336** (0.961)	1.566 (1.809)	−1.291 (1.968)	−1.558 (1.280)	0.267 (0.745)
2014	1.478 (0.894)	0.926* (0.467)	0.552 (0.688)	1.454 (1.565)	−0.035 (0.557)	1.489 (1.073)
2016	−26.457 (16.907)	−2.161 (3.711)	−24.296* (14.180)	1.439 (3.175)	3.034 (3.232)	−1.594 (1.004)
2017	−32.706 (21.394)	−3.925 (5.380)	−28.782* (16.760)	2.881 (4.898)	3.755 (4.144)	−0.873 (1.092)
2018	−36.781 (24.806)	−4.149 (6.238)	−32.632* (19.435)	3.049 (4.753)	4.002 (4.069)	−0.954 (1.163)
2019	−43.706 (28.869)	−6.166 (7.294)	−37.539 (22.460)	2.934 (5.543)	3.781 (4.779)	−0.847 (1.253)
Constant	104.296*** (14.044)	49.285*** (6.758)	55.011*** (13.949)	38.114*** (3.517)	7.757* (4.192)	30.357*** (4.294)
N	1134	1134	1134	1134	1134	1134
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
LZ × year FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.998	0.999	0.995	0.995	0.993	0.995

Notes: Employment outcomes are population estimates, in thousands of workers, constructed using the LALFS city- and county-level weights. Standard errors clustered at the living-zone level are in parentheses. All specifications control for the contemporaneous FTA tariff level, the export-side FTA gap, and year interactions with 2013 routine- and less-educated-employment shares. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 4. Effect of the China–Korea FTA on employment by job type



Notes: Employment outcomes are population estimates, in thousands of workers, constructed using LALFS city- and county-level weights. Points are coefficient estimates from the baseline specification; vertical bars show 90% confidence intervals. The omitted year is 2015.

3.4 By education level

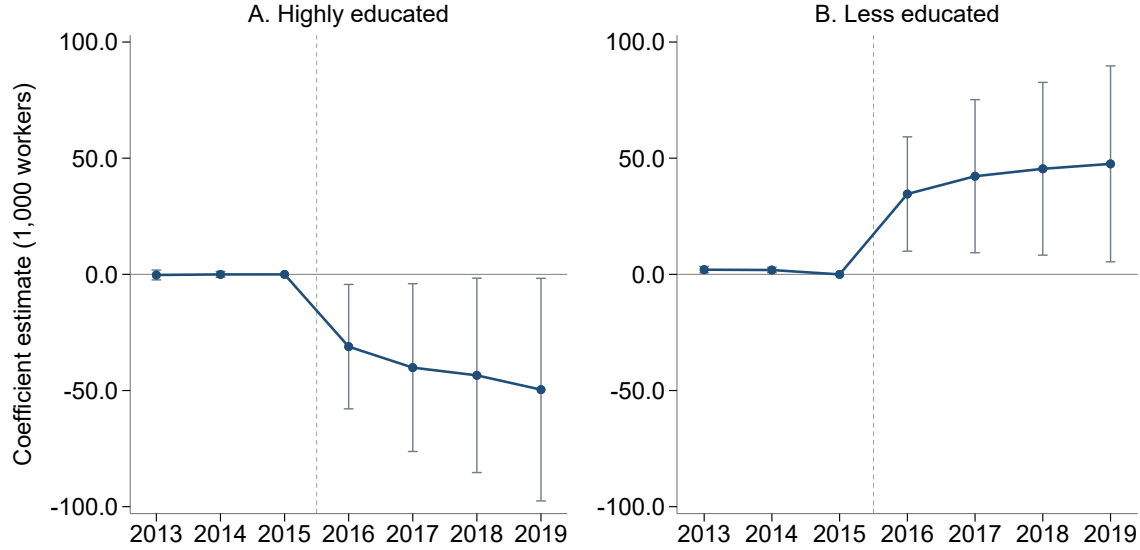
Finally, we analyze whether the China–Korea FTA had different effects on workers with varying levels of education. For example, Borjas and Ramey (1995) found that increased trade had a negative impact on relatively less-educated workers in the United States. To examine whether a similar pattern appears in Korea, we estimate Equation (3) separately for workers with and without a college degree (i.e., highly educated and less educated workers). We find the opposite pattern. Table 5 and Figure 5 show contrasting changes by education. Less-educated employment increases significantly in every post-FTA year for both men and women, while highly-educated employment has negative estimates that are generally significant at the 10% level. This pattern is consistent with the increase in routine employment. However, the significant pre-FTA coefficients for less-educated employment warrant caution in interpreting these estimates causally.

Table 5. Effect of the China–Korea FTA on employment by education level

	Highly educated			Less educated		
	(1) Overall	(2) Male	(3) Female	(4) Overall	(5) Male	(6) Female
2013	−0.273 (1.293)	−0.019 (1.094)	−0.254 (0.462)	2.022** (0.834)	0.702 (0.510)	1.320** (0.558)
2014	−0.016 (0.757)	−0.261 (0.569)	0.245 (0.414)	1.878*** (0.698)	1.067** (0.460)	0.811* (0.458)
2016	−31.116* (16.275)	−8.527* (4.669)	−22.589* (11.860)	34.590** (14.961)	25.367** (12.111)	9.223*** (3.014)
2017	−40.130* (21.958)	−12.854* (7.364)	−27.275* (14.740)	42.246** (20.035)	30.748** (15.132)	11.497** (5.037)
2018	−43.469* (25.434)	−13.211 (8.266)	−30.259* (17.324)	45.441** (22.607)	33.612* (17.092)	11.829** (5.678)
2019	−49.615* (29.118)	−15.874 (9.488)	−33.741* (19.819)	47.563* (25.628)	35.509* (19.261)	12.054* (6.531)
Constant	107.839*** (16.787)	47.436*** (5.522)	60.403*** (13.933)	30.041 (27.107)	9.017 (20.700)	21.024*** (7.014)
N	1134	1134	1134	1134	1134	1134
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
LZ × year FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.998	0.999	0.994	0.999	0.998	0.999

Notes: Employment outcomes are population estimates, in thousands of workers, constructed using the LALFS city- and county-level weights. Standard errors clustered at the living-zone level are in parentheses. All specifications control for the contemporaneous FTA tariff level, the export-side FTA gap, and year interactions with 2013 routine- and less-educated-employment shares. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 5. Effect of the China–Korea FTA on employment by education level



Notes: Employment outcomes are population estimates, in thousands of workers, constructed using LALFS city- and county-level weights. Points are coefficient estimates from the baseline specification; vertical bars show 90% confidence intervals. The omitted year is 2015.

4 Concluding remarks

This paper examines the effects of the China–Korea FTA, which entered into force in December 2015, on the Korean labor market. We find no statistically significant effect on overall employment in regions more exposed to import tariff reductions. The estimates instead point to changes in employment composition: male employment increases while female employment has negative but imprecise estimates; routine and less-educated employment increase, whereas manual and highly-educated employment decrease. We do not find statistically significant changes in overall employment by job type. Because some subgroup outcomes exhibit pre-FTA differences, these heterogeneous effects should be interpreted cautiously. The findings nevertheless suggest that policy responses to the labor market effects of the China–Korea FTA should account for differences in local industrial and labor market structures, as well as differences across gender, occupation, and education groups.

Declaration of interest

No known conflicts of interest associated with this publication could have influenced its outcome.

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References

- Acemoglu, D. and Autor, D. (2011). “Skills, tasks and technologies: Implications for employment and earnings”. In: *Handbook of Labor Economics*. Vol. 4, pp. 1043–1171.
- Acemoglu, D., Autor, D., Dorn, D., Hanson, G. H. and Price, B. (2016). “Import competition and the great US employment sag of the 2000s”. In: *Journal of Labor Economics* 34.S1, S141–S198.
- Aghion, P., Bergeaud, A., Lequien, M., Melitz, M. J. and Zuber, T. (2024). “Opposing firm-level responses to the China shock: Output competition versus input supply”. In: *American Economic Journal: Economic Policy* 16.2, pp. 249–269.
- Autor, D. H., Dorn, D. and Hanson, G. H. (2013). “The China syndrome: Local labor market effects of import competition in the United States”. In: *American Economic Review* 103.6, pp. 2121–2168.
- Autor, D. H., Dorn, D. and Hanson, G. H. (2015). “Untangling trade and technology: Evidence from local labour markets”. In: *The Economic Journal* 125.584, pp. 621–646.
- Besedeš, T., Lee, S. H. and Yang, T. (2021). “Trade liberalization and gender gaps in local labor market outcomes: Dimensions of adjustment in the United States”. In: *Journal of Economic Behavior & Organization* 183, pp. 574–588.
- Borjas, G. J. and Ramey, V. A. (1995). “Foreign competition, market power, and wage inequality”. In: *The Quarterly Journal of Economics* 110.4, pp. 1075–1110.
- Feenstra, R. C., Ma, H. and Xu, Y. (2019). “US exports and employment”. In: *Journal of International Economics* 120, pp. 46–58.
- Feenstra, R. C. and Sasahara, A. (2018). “The “China shock,” exports and US employment: A global input–output analysis”. In: *Review of International Economics* 26.5, pp. 1053–1083.
- Kim, D. (2006). “The impact of Chinese economic growth on the Korean labor market”. In: *The Structural Changes in the Korean Economy after the Economic Crisis*. Ed. by Shin, I. and Han, J. In Korean. KDI.
- Kim, J. and Choi, K. (2015). “The impact of trade expansion with China on Korean labor market”. In: *The Structural Changes in Chinese Economy and Opportunities in Korea*. Ed. by Choi, K. In Korean. KDI.

- Kim, S. H. and Shikher, S. (2015). “Long-run effects of the Korea–China free-trade agreement”. In: *Journal of East Asian Economic Integration* 19.2, pp. 117–142.
- Koo, K. H. and Whang, U. (2018). “The rise of China and the rebound in Korea’s manufacturing employment”. In: *KIEP Research Paper. Working paper*, pp. 18–07.
- Park, Y. J., Shim, M., Yang, H.-S. and Yoo, S. Y. (2023). “Is job polarization path-dependent? Evidence from Korea”. In: *Applied Economics Letters* 30.18, pp. 2495–2499.
- Pierce, J. R. and Schott, P. K. (2016). “The surprisingly swift decline of US manufacturing employment”. In: *American Economic Review* 106.7, pp. 1632–1662.
- Pierce, J. R. and Schott, P. K. (2020). “Trade liberalization and mortality: Evidence from US counties”. In: *American Economic Review: Insights* 2.1, pp. 47–64.
- Shim, M. and Yang, H. S. (2018). “Interindustry wage differentials, technology adoption, and job polarization”. In: *Journal of Economic Behavior & Organization* 146, pp. 141–160.

Appendix

Table A.1. $FTAGAP_i$ across industries, 2016

Two-digit ISIC4 industry	$FTAGAP_i$
Fabricated metal products, except machinery and equipment	4.77703
Other non-metallic mineral products	4.63685
Motor vehicles, trailers, and semi-trailers	3.93647
Textiles, wearing apparel, leather, and related products	3.82288
Chemicals and pharmaceutical products	3.70840
Electrical equipment	3.70141
Food products, beverages, and tobacco	3.36997
Rubber and plastics products	3.28601
Machinery and equipment N.E.C.	3.13201
Agriculture, forestry, and fishing	2.96905
Coke and refined petroleum products	2.72749
Other manufacturing; repair and installation of machinery and equipment	2.27567
Computer, electronic, and optical products	2.22065
Electricity, gas, water supply, sewerage, waste, and remediation services	2
Wood and products of wood and cork (except furniture)	1.73415
Other transport equipment	1.458
Mining and quarrying of non-energy-producing products	1.03142
Other business sector services	0.92857
Manufacture of basic metals	0.83540
Paper products and printing	0.81867
Mining and extraction of energy-producing products	0.46364
Mining support service activities	0
Construction	0
Wholesale and retail trade; repair of motor vehicles	0
Transportation and storage	0
Accommodation and food services	0
Publishing, audiovisual and broadcasting activities	0
Telecommunications	0
IT and other information services	0
Financial and insurance activities	0
Real estate activities	0
Public administration and defense; compulsory social security	0
Education	0

Continued on next page

Table A.1 (continued)

Two-digit ISIC4 industry	$FTAGAP_i$
Human health and social work	0
Arts, entertainment, recreation, and other service activities	0
Private households with employed persons	0